

Key Findings Consistency Audit

Date: 2026-03-21

Scope: README.md vs docs/core/technical_report.md vs recomputed baseline data

Baseline data source: results/baseline/simulation/results_K20.csv (30 replications per policy)

Cross-checked against: results/baseline/tables/table1_baseline_comparison.csv , results/baseline/tables/descriptive_statistics.csv , results/baseline/tables/exp1_summary.csv , results/baseline/simulation/validation_pilot/pilot1_p0_vs_p2.json

1. COMPARISON TABLE

P0 (Spatially-Stratified Baseline) at K=20

Metric	README	Technical Report	Recomputed Baseline	table1_base line	Match?
Mean RT (min)	3.17	3.17	3.165 (CI: 3.142–3.188)	3.17	✓ All consistent (rounding)
P90 RT (min)	(not reported)	5.62	5.334	5.33	✗ Tech report WRONG (5.62 ≠ 5.33)
P95 RT (min)	6.26	(not reported)	6.280	(not in table1)	✓ README matches
6-min Coverage	(not reported)	94.0%	94.0%	(not in table1)	✓ Tech report matches
8-min Coverage	99.6%	99.6%	99.7%	99.7%	⚠ Both docs say 99.6%, baseline says 99.7%
Utilization	7.6%	7.8%	7.57% (≈7.6%)	0.076 (7.6%)	✗ Tech report WRONG (7.8% ≠ 7.6%)

P1 (Demand-Proportional) at K=20

Metric	README	Technical Report	Recomputed Baseline	table1_base line	Match?
Mean RT (min)	(not reported)	2.63	2.619	2.62	⚠️ Tech report rounds to 2.63, table1 to 2.62
P90 RT (min)	(not reported)	4.03	3.988	3.99	✅ Consistent (rounding)
6-min Coverage	(not reported)	98.0%	98.0%	(not in table1)	✅
8-min Coverage	(not reported)	99.6%	99.6%	99.6%	✅
Utilization	(not reported)	7.5%	7.45% (≈7.4%)	0.074 (7.4%)	⚠️ Tech report rounds up to 7.5%

P2 (Demand-Weighted MIP Optimized) at K=20

Metric	README	Technical Report	Recomputed Baseline	table1_base line	Match?
Mean RT (min)	2.57	2.57	2.567	2.57	✅ All consistent
P90 RT (min)	(not reported)	3.76	3.752	3.75	✅ Consistent (rounding)
P95 RT (min)	4.66	(not reported)	4.688	(not in table1)	✅ README matches (rounding)
6-min Coverage	(not reported)	98.2%	98.2%	(not in table1)	✅
8-min Coverage	99.6%	99.6%	99.7%	99.7%	⚠️ Both docs say 99.6%, baseline says 99.7%
Utilization	7.5%	7.5%	7.41% (≈7.4%)	0.074 (7.4%)	⚠️ Both docs round up to 7.5%

Improvement Metrics (P2 vs P0)

Metric	README	Technical Report	Recomputed	Match?
Mean RT improvement	-19.0%	-18.9%	-18.9%	⚠️ README rounds (18.93% → 19.0%)
P90 RT improvement	(N/A)	-33.1%	-29.7%	❌ Tech report WRONG (uses stale P0 P90=5.62)
P95 RT improvement	-25.6%	(N/A)	-25.4%	✅ README close (rounding)

2. CANONICAL CORRECT VALUES (from `results_K20.csv`, $n=30$)

Metric	P0	P1	P2
Mean RT (min)	3.17 [3.14, 3.19]	2.62 [2.60, 2.64]	2.57 [2.55, 2.59]
P90 RT (min)	5.33	3.99	3.75
P95 RT (min)	6.28	5.01	4.69
6-min Coverage	94.0%	98.0%	98.2%
8-min Coverage	99.7%	99.6%	99.7%
Utilization	7.6%	7.4%	7.4%

Improvement (P2 vs P0)	Value
Mean RT	−18.9%
P90 RT	−29.7%
P95 RT	−25.4%
6-min Coverage	+4.2 pp
8-min Coverage	0 pp

3. WHICH DOCUMENT IS WRONG

Technical Report has 3 material errors:

- P0 P90 Response Time = 5.62 min** → Should be **5.33 min**
 - Source of error: Appears to be stale data from an earlier pilot run. The pilot1 JSON shows P0 P90 mean = 5.50 min (also different from 5.62), suggesting the 5.62 value predates even the validation pilot and was never updated when production results were generated.
 - Impact: Cascading error — the “−33.1% P90 improvement” claim is also wrong (should be −29.7%).
- P0 Utilization = 7.8%** → Should be **7.6%**
 - Source of error: Likely from pilot1 or an earlier run with slightly different parameters/seeds. The production baseline consistently shows 7.57% (rounds to 7.6%).
- P0 8-min Coverage = 99.6%** → Should be **99.7%**
 - Minor rounding error. The raw mean across 30 reps is 99.67%, which rounds to 99.7%, not 99.6%.

README has 2 minor issues:

1. **Mean RT improvement = −19.0% → Should be −18.9%**
- Aggressive rounding (18.93% → 19.0%). The tech report correctly says 18.9%.
2. **8-min Coverage = 99.6% for both P0 and P2 → P0 and P2 should both be 99.7%**
- Same rounding issue as the tech report.

Verdict: Technical Report is MORE wrong than README. README is largely correct with minor rounding differences. Technical report has a stale P90 value (5.62) that creates a cascading error in the P90 improvement claim.

4. RECOMMENDED FIX

Technical Report (docs/core/technical_report.md)

Fix the §1 Key Findings table (line ~62-68):

P0 (Spatially-Stratified)	3.17	[3.14, 3.19]	5.33	94.0%	99.7%	7.6%
P1 (Proportional)	2.62	[2.60, 2.64]	3.99	98.0%	99.6%	7.4%
P2 (Optimized)	2.57	[2.55, 2.59]	3.75	98.2%	99.7%	7.4%

Fix the §5.1 table (line ~419-424) — same values as above

Fix improvement claims (lines ~426-428):

- “P2 reduces P90 response time by **29.7%** compared to P0 (from 5.33 to 3.75 min)”
- Keep “−18.9%” for mean RT (already correct)

Fix P0 utilization wherever “7.8%” appears → “7.6%”

README.md

Minor fixes to the Key Findings table (line ~12-17):

- Change “−19.0%” → “−18.9%”
- Change P0 8-min Coverage “99.6%” → “99.7%”
- Change P2 8-min Coverage “99.6%” → “99.7%” (and improvement becomes “0 pp”)

Consistency with table1_baseline_comparison.csv

The table1 CSV is **correct and matches recomputed baseline** for all metrics it reports. It should be treated as the authoritative pre-formatted source.

5. ROOT CAUSE ANALYSIS

The tech report’s stale P90 value (5.62) likely originated from an early simulation run before the production baseline was regenerated. The progression appears to be:

1. **Early pilot run** → P0 P90 ≈ 5.62 (used in tech report draft)
2. **Validation pilot (pilot1_p0_vs_p2.json)** → P0 P90 = 5.50 (improved but never synced to tech report)
3. **Production baseline (results_K20.csv)** → P0 P90 = 5.33 (final canonical value, used in table1)

The tech report was written using values from step 1 and was never fully reconciled with the production results from step 3. The README was updated more recently and uses production-era values, hence fewer discrepancies.